Harshall Relekar

Instrumentation & Automation Engineer

Contact

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Location: Navi Mumbai, Maharashtra

Technical Skills

PLC: Delta PLC, Ladder Logic, I/O Mapping

Protocols: Modbus RTU/TCP, IEC 60870-5-101

Communication: RS485, Coil & Register Mapping

Embedded: ESP32, Embedded C/C++

Automation: Servo Integration, Commissioning

Instrumentation: DP Transmitter Calibration

Debugging: Protocol Analysis & Troubleshooting

RTU: Emerson FB Series, ControlWave, RTU Configuration, System Testing

Education

PG Diploma in Industrial Automation

Diploma in Instrumentation

SSC – Sacred Heart Boy's High School

Harshall Relekar

Automation & Embedded Systems Engineer

Professional Summary

Automation and Instrumentation Engineer with hands-on experience in PLC programming, embedded system development, and industrial communication protocol implementation. Skilled in integrating Modbus RTU/TCP and IEC 60870-5-101 over RS485 networks. Experienced in SCADA monitoring systems, servo motor integration, industrial IoT solutions, and communication-level troubleshooting.

Work Experience

Project Engineer – TECHNIX ACS(EMERSON)

Dec 2025 – Present

- Performed comprehensive testing of Emerson Flow Computers (FB series) and ControlWave RTUs.
- Configured and commissioned RTUs for FB and ControlWave applications as per project requirements.
- Executed Factory Acceptance Testing (FAT) and validation to ensure system compliance and reliability.
- Handled system installation, setup, and on-site integration of control systems.
- Conducted troubleshooting, diagnostics, and performance optimization of field devices.
- Managed client interactions, providing technical support and ensuring successful project delivery.
- Configured and tested communication protocols such as Modbus and TCP/IP.

Project Engineer – Wholestand Enterprises

July 2025 – Dec 2025

- Designed and implemented PLC-based automation systems.
- Integrated servo motors with industrial control logic.
- Developed and commissioned sticker applying automation machine.
- Configured Modbus communication and device scanning.
- Performed system diagnostics and performance optimization.

Program Developer – Ceyone Solutions Pvt. Ltd

March 2024 – June 2025

- Developed industrial automation solutions using Delta PLC.
- Implemented PLC-controlled robotic mechanism.
- Built IoT-based ultrasonic water level monitoring with SMS alerts.
- Designed CT-based electrical monitoring system.
- Integrated surge counter and remote monitoring modules.
- Worked on industrial communication protocol integration and testing.

Calibration Engineer – Baldota Control & Equipment

Jan 2023 – Sept 2023

- Performed calibration and validation of industrial instruments.
- Conducted field accuracy testing of DP transmitters.
- Prepared compliance and calibration reports.
- Troubleshot instrumentation issues in industrial environments.

College Self Projects

Digital Clock using ATmega Microcontroller

- Designed and programmed ATmega-based digital clock system.
- Implemented timer-based logic and hardware interfacing.
- Tested circuit stability and time accuracy.

Analog Temperature Controller using XBee

- Developed wireless temperature monitoring system using XBee module.
- Integrated sensor input with microcontroller-based control logic.
- Tested remote communication reliability.